

PAPER_162 — Solar Cycle UQFF: $\omega_c = 2\pi/(11\text{yr})$, Time-Varying $B(t)$, δ_{def} Defect Factor

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Abstract

This paper establishes **per-body cycle frequency** ω_c as a new first-class UQFF parameter, replacing the static magnetic field B_s with a time-varying field $B(t)$ in the Ug1 magnetic dipole term. The 11-year solar cycle drives $B(t) = B_s + 0.4 \cdot \sin(\omega_c \cdot t)$ for the Sun, while each planet uses its own orbital/rotation period. A new Ug1 **defect factor** $\delta_{\text{def}} = 0.01$ modulates the magnetic dipole oscillation amplitude. This paper is the theoretical foundation for PAPER_157's per-body ω_c parameters.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Motivation

Previous UQFF implementations used static magnetic field B_s . The Sun's magnetic field varies by $\pm 40\%$ over the 11-year solar cycle (B range: $\sim 0.6 \text{ G}$ to $\sim 2 \text{ G}$ at solar average). This time variation affects: - Ug1 (magnetic dipole term) via $\mu_s(t) = B(t) \cdot R_s^3$ - Ug3 (string rotation) via $\omega'_s(t) = \omega_s + \omega_c \cdot \cos(\omega_c \cdot t)$ - Solar wind velocity $\rightarrow \delta_{\text{sw}}$ term in Ug2

2. Time-Varying Magnetic Field

$$B(t) = B_s + 0.4 \cdot \sin(\omega_c \cdot t) + S_{Cm,contrib}$$

where $S_{Cm,contrib}$ is the superconductive medium contribution (perturbative, typically $\sim B_s/100$).

2.1 Magnetic Dipole Moment $\mu_s(t)$

$$\mu_s(t) = B(t) \cdot R_s^3$$

Body	B_s [T]	$\Delta B/B_s$	Period	ω_c [rad/s]
Sun	1×10^{-4}	40%	11 yr	$2\pi/(11 \cdot 365.25 \cdot 86400)$
Earth	3×10^{-5}	varies	1 yr (proxy)	$2\pi/(1 \cdot 365.25 \cdot 86400)$
Jupiter	4×10^{-4}	varies	11.86 yr	$2\pi/(11.86 \cdot 365.25 \cdot 86400)$
Neptune	1×10^{-4}	varies	164.8 yr	$2\pi/(164.8 \cdot 365.25 \cdot 86400)$

3. Modified Ug1 with Defect Factor

$$U_{g1}(r, t) = k_1 \cdot \mu_s(t) \cdot \nabla \frac{M_s}{r} \cdot e^{-\alpha t} \cos(\pi t_n) \cdot (1 + \delta_{def} \cdot \sin(0.001t))$$

where the **defect factor** $\delta_{def} = 0.01$ introduces a slow perturbation at 0.001 rad/s (~6.3 second period) representing surface magnetic flux tube defects.

Physical basis for δ_{def} : Magnetic flux tube emergence/submergence at the solar surface creates ~1% modulation in the effective dipole moment on timescales of seconds to minutes (observed in solar magnetogram data).

4. Modified Ug3 with Time-Varying Rotation

$$\omega'_s(t) = \omega_s + \omega_c \cdot \cos(\omega_c \cdot t)$$

The rotation frequency itself becomes time-modulated by the solar cycle, affecting:

$$U_{g3}(r, t) = k_3 \cdot B_j(t) \cdot \cos(\omega'_s(t) \cdot t \cdot \pi) \cdot P_{core} \cdot E_{react}$$

5. Per-Body ω_c Implementation (C++)

```
// omega_c assignments (radians per second)
const double YEAR = 365.25 * 86400.0;

body_sun.omega_c      = 2.0 * M_PI / (11.0 * YEAR); // 11-year solar cycle
body_earth.omega_c     = 2.0 * M_PI / (1.0 * YEAR); // annual orbital
body_jupiter.omega_c  = 2.0 * M_PI / (11.86 * YEAR); // 11.86-year orbital
body_neptune.omega_c  = 2.0 * M_PI / (164.8 * YEAR); // 164.8-year orbital

// Compute time-varying B
double compute_B_t(const CelestialBody& body, double t) {
    double SCm_contrib = body.SCm_density * 1e-10; // perturbative small
    return body.Bs_avg + 0.4 * std::sin(body.omega_c * t) + SCm_contrib;
}
```

6. Testable Predictions

At solar maximum ($B_s + 0.4 = 1.4 \times 10^{-4}$ T vs minimum $B_s - 0.4 = 0.6 \times 10^{-4}$ T):

$$\frac{U_{g1}^{max}}{U_{g1}^{min}} = \frac{1.4}{0.6} \approx 2.33$$

Solar UQFF field strength varies by factor ~2.3 over the solar cycle. This 11-year modulation should correlate with: - Cosmic ray flux variation (observed: ~10-20%) - Interplanetary field modulation (observed) - Long-term climate variation (Maunder Minimum connection)

7. CP Integration

CP2 update: UQFFSolarCycleCalculator — add omega_c parameter, compute_B_t(), delta_def parameter, modified Ug1 and Ug3 equations with time-varying components.

Status: □ Complete | **CP Stage:** CP2 **Supersedes:** N/A (extends static B_s) | **Related:** PAPER_157 (per-body ω_c usage), PAPER_027 (5-freq resonance including solar), PAPER_086 (Ug1 derivation)

UQFF computed: Solar wind UQFF correction = $[SSq]\exp(-r/v) = 5.7e-1\exp(-5.0e-4(1AU/400km/s)) = 5.7e-1\exp(-3.2e-3) = 5.7e-1$; dominant at $r < 1AU$.